VICTOR NDUNDA

AI Systems Engineer - Multi-Agent AI | RAG | Production ML

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PROFESSIONAL SUMMARY

AI systems engineer who designs and ships production AI - multi-agent platforms, RAG systems, and applied machine learning - for African and global markets, with a bias for systems that survive real-world constraints: low bandwidth, low-end devices, and multilingual users. Built the AI layer of **Tapi Learn** for ACLA, serving 11,000+ learners across 38 countries (EN-FR-SW, 4.8/5 client-reported), and **Keja AI** for CHACADOM INVESTMENTS, a 7-stage real-estate platform in public trial across 12 Kenyan counties. Creator of **Busara AI** (6-stage multi-agent DAG pipeline with statistics implemented from scratch), **KilimoPRO** (offline-first agri-AI for 8 IGAD countries, on-device TensorFlow Lite detection), and **AgentReplay** (deterministic replay for AI agents: 0.67% overhead, 100% reproduction fidelity, MIT-licensed). 5+ years shipping Go, TypeScript, Python, and Flutter systems, on a data-science foundation built inside Kenyan financial institutions (RBA, NSSF, Enwealth). Based in Nairobi - available for remote engagements worldwide.

WORK EXPERIENCE

AI Systems Engineer & Consultant

Independent practice | 2022 - Present | Nairobi, Kenya - remote worldwide

- **Tapi Learn - client engagement for ACLA (2026):** built the 24/7 AI Career Coach, the EN-FR-SW multilingual NLP pipeline, and personalized learning recommendations - 11,000+ learners in 38 countries, 4.8/5 average rating (client-reported).
- **Keja AI - client engagement for CHACADOM INVESTMENTS (2026):** built the "Ask Keja AI" assistant, per-listing investment scoring, and title & document verification for a 7-stage real-estate platform - 87+ listings across 12 Kenyan counties, in public trial.
- **FirstLight (Oct 2026):** self-initiated Geo-AI disaster-response demo - H3 need-score zoning, Dijkstra routing, hospital isochrones, human-in-the-loop review, hash-verified sitreps.
- Delivery: needs assessment through production deployment and team training - from $2,500 to $35,000+ programs.

Creator & Maintainer

AgentReplay (open source) | 2026 - Present | github.com/gadda00/agentreplay

- Deterministic replay and counterfactual debugging layer for AI agents: records every LLM call, tool response, clock read, and RNG draw, then replays the run bit-exact with zero model calls.
- 0.67% recording overhead (target 5%; incumbents 0-15%), 100% reproduction fidelity on the validation set, 99 tests passing; LangGraph-first, adapters for OpenAI, Anthropic, CrewAI, and AutoGen; published a 22-minute research whitepaper.

Creator & Lead AI Engineer

Busara AI | 2023 - Present | busaraai.com

- Multi-agent data intelligence platform: 6-stage parallel DAG pipeline (Ingest, Engineer, Detect, Forecast, Infer, Report) with circuit breakers and smart caching, serving live users; cut average analysis latency by ~40%.
- Implemented statistics from scratch - Holt-Winters, Z-score/IQR/EWMA anomaly ensembles, OLS regression, Granger causality, K-Means++ - and a multi-LLM council (GLM-4.6, GPT, Claude, Gemini) with citation-grounded RAG.

Creator & Lead Engineer

KilimoPRO (open source) | 2022 - Present | github.com/gadda00/kilimopro

- Offline-first agri-AI for 8 IGAD countries: on-device crop disease detection (TensorFlow Lite) for 15 crops on $80 phones; GARCH/VaR market forecasting; multi-LLM advisory council in Swahili and English; 22+ data sources (KilimoSTAT, FAOSTAT, Earth Engine, KALRO); SMS/USSD channels for feature phones; aligned with Kenya's NASIP 2026-2030 strategy.

Research Intern

Retirement Benefits Authority (RBA) | Apr - May 2024 | Nairobi, Kenya

- Researched retirement-industry trends and regulatory frameworks; collected, cleaned, and analyzed large datasets to support evidence-based policy recommendations.
- Prepared detailed reports and visualizations for senior management; research underpins a published paper on AI and machine learning in the retirement industry.

Data Analyst Intern

Enwealth Financial Services | Feb - Jun 2022 | Nairobi, Kenya

- Executed ETL processes for financial datasets; built interactive dashboards and BI reports tracking key metrics; translated statistical trend analysis into actionable recommendations for financial advisors.

Data Scientist & Analytics Engineer

Independent | 2019 - 2022 | Nairobi, Kenya

- Delivered financial forecasting systems (ARIMA, Holt-Winters, exponential smoothing, backtesting) for retirement-benefits and fintech clients - 87.3% mean accuracy across 12 financial metrics.
- Built a Go fraud-detection microservice (Z-Score + IQR + Velocity weighted-vote ensemble, 50-300µs scoring) and customer-segmentation engines (RFM, K-Means, GMM - 23% lift in campaign response).

Administration Intern

National Social Security Fund (NSSF) | Oct 2020 - Jun 2021 | Nairobi, Kenya

- Processed and analyzed benefits applications in a regulated pension environment, ensuring compliance with statutory requirements; used data analysis to optimize processing workflows and managed sensitive member data under strict security protocols.

SELECTED PROJECTS & OPEN SOURCE

- **Keja AI** (keja.app) - real-estate intelligence & trust infrastructure: 7-stage lifecycle, 8 stakeholder workspaces, 87+ listings in public trial.
- **Tapi Learn** (acla.io/tapi-learn) - multilingual AI learning platform for ACLA: 11K+ learners, 38 countries, 4.8/5 (client-reported).
- **FirstLight** (gadda00.github.io/firstlight) - Geo-AI disaster-response decision layer; public demo, provenance-labeled.
- **Verxlite** (github.com/gadda00/verxlite) - AI workflow agent for sales & ops; public 281-defect code review, full P0/P1 remediation, 83 backend tests.
- **Fraud Detection System** (github.com/gadda00/fraud-detection-system) - Go engine with 7 statistical detectors: 84% recall, 1.37% FPR, ~33µs per score; PCI-DSS / PSD2 / GDPR Article 22-aware.
- **IntelliFlow** (github.com/gadda00/IntelliFlow) - 33-agent self-evolving multi-agent platform; the open-source foundation of Busara AI.
- **Eco Africa Awards** (github.com/gadda00/eco-africa-awards) - awards platform for African climate leadership, an ACLA initiative; Next.js, Three.js, AI.
- **Hazina Assist** - new AI platform in active development; public details at the first milestone (2027).

RESEARCH & PUBLICATIONS

- **The Transformative Role of AI and Machine Learning in Modernizing the Retirement Industry** - grounded in research conducted at the Retirement Benefits Authority. tinyurl.com/researchvmn
- **Advanced Data Flow Visualization and Governance in Complex Enterprise Systems** - tinyurl.com/researchvmn2

EDUCATION

B.Sc. in Computer Science

University of Embu | 2018 - 2024 | Embu, Kenya

- Coursework: Data Structures & Algorithms, Database Management Systems, Statistical Methods, Software Engineering, Network Security, Machine Learning. Final-year project: an early crop advisory system that seeded the KilimoPRO concept.

TECHNICAL SKILLS

Languages: Python, Go, TypeScript/JavaScript, Dart, SQL, Bash

LLMs & agents: GPT, Claude, Gemini, GLM-4.6, Kimi K2, Mistral, Qwen, DeepSeek, Llama; LangChain, LangGraph, LlamaIndex, CrewAI, AutoGen; custom multi-agent DAG orchestration

RAG & ML: pgvector, Pinecone, Weaviate, hybrid search, reranking; TensorFlow / TensorFlow Lite, scikit-learn, Holt-Winters, GARCH/VaR, K-Means++, Granger causality

Web & infra: Next.js, React, Node.js, FastAPI, Gin, Flutter, Celery; PostgreSQL, Redis, Docker, Kubernetes, GCP (Earth Engine), Prometheus, OpenTelemetry, CI/CD

Data & analytics: pandas, NumPy, Advanced Excel (Power Query, DAX), ETL processes, BI dashboards, data modeling, trend & statistical analysis

CERTIFICATIONS & LANGUAGES

- **AI Engineer for Data Scientists Associate** and **AI Engineer for Developers Associate** - in progress
- **Languages:** English (professional) | Kiswahili (native) | German (basic) | French (basic) - references available on request.